Gopika Premsankar

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ORCID: 0000-0003-3463-6077

Education

2015 – 2020 **Doctor of Science (Tech.)**, Department of Computer Science, *Aalto University*, Finland

Supervisor: Prof. Mario Di Francesco, *advisor*: Prof. Tarik Taleb

Dissertation: Scalable networked systems: analysis and optimization

2013 – 2015 **Master of Science (Tech.) with distinction**, Department of Computer Science, *Aalto University*, Finland

Graduated with distinction

Thesis: Design and implementation of a distributed MME on OpenStack

2006 – 2010 **Bachelor of Technology**, Department of Electrical and Electronics Engineering, *National Institute of Technology Karnataka*, Surathkal, India

Professional experience

Jan '24 – **Assistant professor, tenure-track**, *Aalto University*, Espoo, Finland.
Department of Information and Communications Engineering,
School of Electrical Engineering

Mar '23 **Visiting researcher**, *VU Amsterdam*, Amsterdam, Netherlands.
Project: One-month research visit on carbon-aware networking and edge computing
Host: Prof. Lin Wang

Sep '21 – Jan '24 **Postdoctoral researcher**, *University of Helsinki*, Helsinki, Finland.
Project: Scalable and energy-efficient networked systems at the edge
Supervisor: Prof. Sasu Tarkoma

Feb '21 – Aug '21 **Postdoctoral researcher**, *Ivey Business School, Western University*, London, ON, Canada.
Project: Optimization of energy in edge networks
Supervisor: Prof. Bissan Ghaddar

Apr '20 – Jan '21 **Postdoctoral researcher**, *Aalto University*, Espoo, Finland.
Project: Modeling LoRa networks
Supervisor: Prof. Mario Di Francesco

Sep – Dec '19 **Research intern**, *Nokia Bell Labs*, Dublin, Ireland.
Project: Data-driven approach to improve energy efficiency of cellular base stations
Supervisor: Dr. Diego Lugones

Jun – Sep '19 **Visiting PhD student**, *Duke University*, Durham, NC, USA.
Project: Investigation of networking challenges for augmented reality applications
Host: Prof. Maria Gorlatova

May '18 **Visiting PhD student**, *National Chiao Tung University*, Hsinchu, Taiwan.
Two week visit to exchange ideas related to edge computing and Internet of Things
Host: Prof. Yu-Chee Tseng

May – Sep '16 **Research intern**, *IBM Research*, Dublin, Ireland.
Project: Edge computing for vehicular applications in smart cities
Host: Prof. Bissan Ghaddar

2011 – 2013 **Software engineer**, *Cisco Systems India Private Limited*, Bangalore, India.
Test engineer for mobile packet core network elements
Developed deep understanding of LTE networks and telecommunications industry

2010 – 2011 **Associate software engineer**, *Accenture*, Bangalore, India.
Application developer in pharmaceutical and life sciences division

Research funding

Sep '25 – Aug '29 **DecAI: Decarbonizing AI through Long-term Impact Assessment and Optimization**, Research Council of Finland Academy Research Fellowships 2025, Principal investigator, Award amount: 627 692 EUR

The project aims to estimate and reduce carbon emissions across the entire lifecycle of AI, from development to operation and use. It also designs optimization algorithms that manage computing resources in a carbon-efficient way, supporting both sustainability goals and long-term benefits for the electricity grid.

Sep '21 – Jan '24 **Scalable and Energy-efficient Networked Systems at the Edge**, Academy of Finland postdoctoral researcher, Award amount: 240,820 EUR

The project establishes energy-efficiency as a fundamental metric in designing and deploying applications in edge data centers. The goal is to minimize the energy consumed of networking infrastructure while still supporting the always-on, connected services of the future. The project ended in Jan '24 on starting as faculty at Aalto University.

Feb '21 – Jan '22 **Optimization of energy in edge networks**, postdoc pooli (Finnish Cultural Foundation), Principal investigator, Award amount: 52,000 EUR

The project involved devising new optimization models to allocate resources in data centers such that energy consumed is minimized. The project ended in Aug '21 as I received other funding.

Publications

Journals

[J9] D. Xu, X. Su, G. Premsankar, H. Wang, S. Tarkoma, and P. Hui. Dynamic Hierarchical Reinforcement Learning Framework for Energy-Efficient 5G Base Stations in Urban Environments. In: *IEEE Transactions on Mobile Computing* (2025), pp. 1–18. DOI: https://doi.org/10.1109/TMC.2025.3557280.

[J8] J. Jeong, G. Premsankar, B. Ghaddar, and S. Tarkoma. A Robust Optimization Approach for Placement of Applications in Edge Computing considering Latency Uncertainty. In: *Omega* (2024), p. 103064. DOI: https://doi.org/10.1016/j.omega.2024.103064.

[J7] G. Premsankar and B. Ghaddar. Energy-Efficient Service Placement for Latency-Sensitive Applications in Edge Computing. In: *IEEE Internet of Things Journal* (2022). DOI: 10.1109/JIOT.2022.3162581.

[J6] V. Toro-Betancur, G. Premsankar, C.-F. Liu, M. Słabicki, M. Bennis, and M. Di Francesco. Learning How to Configure LoRa Networks with No Regret: a

Distributed Approach. In: *IEEE Transactions on Industrial Informatics* (2022). DOI: 10.1109/TII.2022.3187721.

[J5] Y. Zhang, T. Scargill, A. Vaishnav, G. Premsankar, M. Di Francesco, and M. Gorlatova. InDepth: Real-time Depth Inpainting for Mobile Augmented Reality. In: *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 6.1 (Mar. 2022). DOI: 10.1145/3517260.

[J4] G. Premsankar*, G. Piao*, P. K. Nicholson, M. D. Francesco, and D. Lugones. Data-Driven Energy Conservation in Cellular Networks: A Systems Approach. In: *IEEE Transactions on Network and Service Management* 18.3 (2021), pp. 3567–3582. DOI: 10.1109/TNSM.2021.3083073. *Equal contribution.

[J3] B. Jedari, G. Premsankar, G. Illahi, M. Di Francesco, A. Mehrabi, and A. Ylä-Jääski. Video Caching, Analytics and Delivery at the Wireless Edge: A Survey and Future Directions. In: *IEEE Communications Surveys & Tutorials* 23.1 (2021), pp. 431–471. DOI: 10.1109/COMST.2020.3035427.

[J2] G. Premsankar, B. Ghaddar, M. Slabicki, and M. Di Francesco. Optimal configuration of LoRa networks in smart cities. In: *IEEE Transactions on Industrial Informatics* (2020). DOI: 10.1109/TII.2020.2967123.

[J1] G. Premsankar, M. Di Francesco, and T. Taleb. Edge computing for the Internet of Things: A case study. In: *IEEE Internet of Things Journal* 5.2 (2018), pp. 1275–1284. DOI: 10.1109/JIOT.2018.2805263.

Book chapter

[B2] G. Premsankar and S.-M. Laaksonen. Estimating the carbon footprint of digital services. In: *Platforms and the Planet : Big Tech, Digital Platforms and Environmental Responsibility.* In press. Emerald Publishing Limited, 2025.

[B1] G. Premsankar and M. Di Francesco. Advances in Cloud Computing, Wireless Communications and the Internet of Things. In: *Analytics for the Sharing Economy: Mathematics, Engineering and Business Perspectives.* Springer, 2020, pp. 71–94. DOI: 10.1007/978-3-030-35032-1_6.

Conferences

[C12] M. Ghafoori, G. Premsankar, and X. Tu. Are We Measuring What Matters? Reframing Evaluation of Extended Reality for Industry 5.0 Work Environments. In: *First International Workshop on Working Enhancement with Extended Reality.* Oct. 2025. To appear.

[C11] V. Toro-Betancur, G. Premsankar, L. Corneo, and M. Di Francesco. Timely Data Delivery for Heterogeneous IoT Applications. In: *2025 23rd International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt).* IEEE. 2025, pp. 22–29. DOI: 10.23919/WiOpt66569.2025.11123281.

[C10] V. Toro-Betancur, G. Premsankar, L. Corneo, and M. Di Francesco. Timely Data Delivery for Heterogeneous IoT Applications. In: *Proceedings of the 23rd International Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks.* 2025. To appear.

[C9] M. Lintunen, G. Premsankar, H. Tenhunen, S. Tarkoma, and A. Rao. Poster: Automatic Mass Power Outage Detection in Radio Access Networks. In: *Proceedings of the 21st Annual International Conference on Mobile Systems, Applications and Services*. 2023, pp. 559–560. DOI: 10.1145/3581791.3597365.

[C8] T. Scargill, G. Premsankar, J. Chen, and M. Gorlatova. Here To Stay: A Quantitative Comparison of Virtual Object Stability in Markerless Mobile AR. In: *Second International Workshop on Cyber-Physical-Human System Design and Implementation*. May 2022, pp. 24–29. DOI: 10.1109/CPHS56133.2022.9804545.

[C7] V. T. Betancur, G. Premsankar, M. Slabicki, and M. Di Francesco. Modeling communication reliability in LoRa networks with device-level accuracy. In: *INFOCOM 2021-IEEE Conference on Computer Communications*. IEEE. 2021. DOI: 10.1109/INFOCOM42981.2021.9488783. **Acceptance rate: 19.9%**.

[C6] G. Premsankar, B. Ghaddar, M. Di Francesco, and R. Verago. Efficient placement of edge computing devices for vehicular applications in smart cities. In: *NOMS 2018-2018 IEEE/IFIP Network Operations and Management Symposium*. IEEE. 2018, pp. 1–9. DOI: 10.1109/NOMS.2018.8406256. **Best student paper award**.

[C5] S. K. Mohanty, G. Premsankar, and M. Di Francesco. An Evaluation of Open Source Serverless Computing Frameworks. In: *CloudCom*. 2018, pp. 115–120. DOI: 10.1109/CloudCom2018.2018.00033.

[C4] M. Slabicki, G. Premsankar, and M. Di Francesco. Adaptive configuration of LoRa networks for dense IoT deployments. In: *NOMS 2018-2018 IEEE/IFIP Network Operations and Management Symposium*. IEEE. 2018, pp. 1–9. DOI: 10.1109/NOMS.2018.8406255.

[C3] S. Bayhan, G. Premsankar, M. Di Francesco, and J. Kangasharju. Mobile content offloading in database-assisted white space networks. In: *International Conference on Cognitive Radio Oriented Wireless Networks*. Springer. 2016, pp. 129–141. DOI: 10.1007/978-3-319-40352-6_11.

[C2] G. Premsankar, K. Ahokas, and S. Luukkainen. Design and implementation of a distributed mobility management entity on OpenStack. In: *2015 IEEE 7th International Conference on Cloud Computing Technology and Science (CloudCom)*. IEEE. 2015, pp. 487–490. DOI: 10.1109/CloudCom.2015.54. Short paper.

[C1] J. Costa-Requena, J. L. Santos, V. F. Guasch, K. Ahokas, G. Premsankar, S. Luukkainen, O. L. Pérez, M. U. Itzazelaia, I. Ahmad, M. Liyanage, et al. SDN and NFV integration in generalized mobile network architecture. In: *2015 European conference on networks and communications (EuCNC)*. IEEE. 2015, pp. 154–158. DOI: 10.1109/EuCNC.2015.7194059.

Talks, presentations and scientific meetings

29/09-02/10/2024 Invited attendee at the Dagstuhl seminar on Greening Networking: Toward a Net Zero Internet

24/03/2023	Scalable and energy-efficient networked systems VU Amstersdam, *Amsterdam*, Netherlands
13/03/2023	Scalable and energy-efficient networked systems University of Twente, *Enschede*, Netherlands
06/07/2022	Energy-Efficient Service Placement for Latency-Sensitive Applications in Edge Computing, 32nd European Conference on Operational Research (EURO 2022), *Espoo*, Finland
12/10/2021	Invited (virtual) guest lecture: LoRa, Autumn 2021: Networked Systems and Services, *University of Helsinki*, Finland
22/03/2021	Invited (virtual) guest lecture: Infrastructure for edge computing, Spring 2021: Edge Computing, *Duke University*, NC, USA
05/08/2019	Elevator pitch: towards scalable communication networks, NeTS Early Career Work shop 2019, *National Science Foundation*, Arlington, VA, USA
11/10/2018	Demo: long range connectivity for IoT, Research day, *Aalto University*, Finland
10/05/2018	Research directions for edge computing and LoRa, *National Chiao Tung University*, Hsinchu, Taiwan
25/04/2018	Adaptive configuration of LoRa networks for dense IoT deployments, Efficient placement of edge computing devices for vehicular applications in smart cities, conference paper presentations, *IEEE NOMS 2018*, Taipei, Taiwan
07/09/2016	Optimizing the placement of edge devices in vehicular networks, final internship presentation, *IBM Research*, Dublin, Ireland
29/06/2016	Opportunistic content offloading using white space and ISM spectrum, *IBM Research*, Dublin, Ireland
03/12/2015	Design and implementation of a distributed Mobility Management Entity on OpenStack, conference paper presentation, *IEEE CloudCom 2015*, Vancouver, Canada

Awards and honours

Aug '19	Travel grant for NeTS Early Career Workshop, National Science Foundation, VA, USA
Spring '19	Aalto Foundation travel grant for research visit to Duke University
Apr '18	Best student paper award, IEEE/IFIP NOMS 2018
Apr '18	Student travel grant, IEEE/IFIP NOMS 2018, Apr 23-27, 2018, Taipei, Taiwan
Nov '15	Student travel grant, IEEE CloudCom, Nov 30-Dec 3, 2015, Vancouver, Canada
2013 – 2015	Aalto University Category B Scholarship for Master's study programme
2012, 2013	Two Cisco Achievement Program awards for excellent work
2010, 2011	Quarterly awards for "Excellence as a Business Operator" at Accenture
2006 – 2010	Scholarship for Bachelor's study programme, Scholarship Programme for Diaspora Children (Ministry of Overseas Indian Affairs, Government of India)

Teaching and supervision

Doctoral researcher supervisor

Jun '24 -	César Olvera Espinosa
Aug '24 -	Antti Heikinmäki (part-time industry doctoral researcher)
Apr - Aug '25	Ali Pashazadeh (visiting doctoral researcher), supervisors: Giovanni Stea, Giovanni Nardini

Master's thesis supervision

Ongoing	3 M.Sc. thesis students
2018 – 2025	Supervised 6 M.Sc. thesis students at Aalto University, advised 2 M.Sc. thesis students at University of Helsinki, advised 6 M.Sc. thesis students at Aalto University

Bachelor's thesis supervisor

Ongoing	1
2024-2025	Supervised 3 B.Sc. thesis students at Aalto University

Teaching

Spring '25, '26	Responsible teacher for Networking at Scale and Advanced Applications
Mar '25	Guest lecture: Sustainability of AI, Spring 2025: Elements of Sustainable ICT, Aalto University, Finland
Feb '24	Guest lecture: Sustainability of AI, Spring 2024: Elements of Sustainable ICT, Aalto University, Finland
Summer '23	Guest lecture for Summer school on communications engineering and data science, Aalto University

Teaching assistant

Fall '22	DATA11003: Distributed Data Infrastructures, University of Helsinki
Fall '21	CSM13001: Distributed Systems, University of Helsinki
Fall '20	CS-E4190: Cloud Software and Systems, Aalto University
Fall '17, '18	CS-E4100: Mobile Cloud Computing, Aalto University
Spring '17	CS-E4002: The Internet of Things: Selected Themes, Aalto University
Fall '14, '15, '16	CS-E4005 Methods and Tools for Network Systems, Aalto University

Pedagogical training

Spring '24	Learning and teaching in higher education (Aalto University)
Autumn '23	Constructive alignment in course design (University of Helsinki)
Spring '20	A! Peda Intro (Aalto University)

Academic service

Technical program committee member

2025	Tenth ACM/IEEE Symposium on Edge Computing (IEEE/ACM SEC 2025)
2025	23rd ACM Conference on Embedded Networked Sensor Systems (ACM SenSys 2025)

2024 International Workshop on Edge Systems, Analytics and Networking (EdgeSys, co-located with EuroSys), International Workshop on Networked AI Systems (NetAISys, co-located with ACM MobiSys), EAI MobiQuitous

2023 International Workshop on Networked AI Systems (NetAISys, co-located with ACM MobiSys)

2019 Grace Hopper Conference for Women in Computing, Computer Systems Engineering (CSE) track committee member

2018 ACM EuroSys (Shadow PC)

Conference organization

2025 **Posters and demos chair**, ACM UbiComp 2025

2024 **Session chair** for session on intelligent performance, IoT Conference 2024 in Oulu, Finland

2024 **Workshop and tutorial chair**, IoT Conference 2024

2023 **Registration chair**, ACM MobiSys 2023

2022 **Session organizer and chair, Applications of OR in telecommunications** for EURO 2022

Reviewer

2025- Distinguished reviewer board member, ACM Transactions on Internet of Things

2016- **Journals:** ACM Computing Surveys, ACM Transactions on Sensor Networks, IEEE Transactions on Mobile Computing, IEEE Transactions on Communications, IEEE Transactions on Parallel and Distributed Systems, IEEE Transactions on Wireless Communications, IEEE Communications Magazine, IEEE Internet of Things Journal, IEEE Systems Journal, IEEE Network Magazine, IEEE Open Journal of the Communications Society, IEEE Journal of Selected Topics in Signal Processing, IEEE Transactions on Sustainable Computing, IEEE Wireless Communications Magazine, IEEE Communications Standards Magazine, Elsevier Pervasive and Mobile Computing, Elsevier Computer Networks, Springer Wireless Networks

Conferences: ICIS 2022, ACM IMWUT / UbiComp 2022, IEEE ICDCS 2021, IEEE WoWMoM (2019, 2020), IEEE Sarnoff 2019, IEEE SMARTCOMP 2017, IEEE PerCom (2016, 2021)

Other

2023 Contributor to massive open online course on 5G: Core 5G and Beyond MOOC

Media coverage

2022 Tammenlastuja (Magazine of the Finnish Cultural Foundation) 03/22, Energiatehokkuus on tulevaisuuden mittari

Additional activities

09/12/2024 Panelist on School of Electrical Engineering's EDI Event, Women's Lunch, *Aalto University*, Finland

27/02/2024 Panelist on "How to catalyze diversity in science?", IUPAC Global Women's Breakfast 2024, *Aalto University*, Finland

2019 – 2020 Website redesign co-chair, board member, N2Women

2017 – 2020 Contributor to open source simulator, Framework for LoRa (FLoRa) for end-to-end simulations of LoRa networks

2017 – 2020 Volunteer in codebar, Women for Women Workshops and Django Girls, Helsinki with the goal to improve representation of underrepresented groups in technology